

# Week3\_Assignment\_TG

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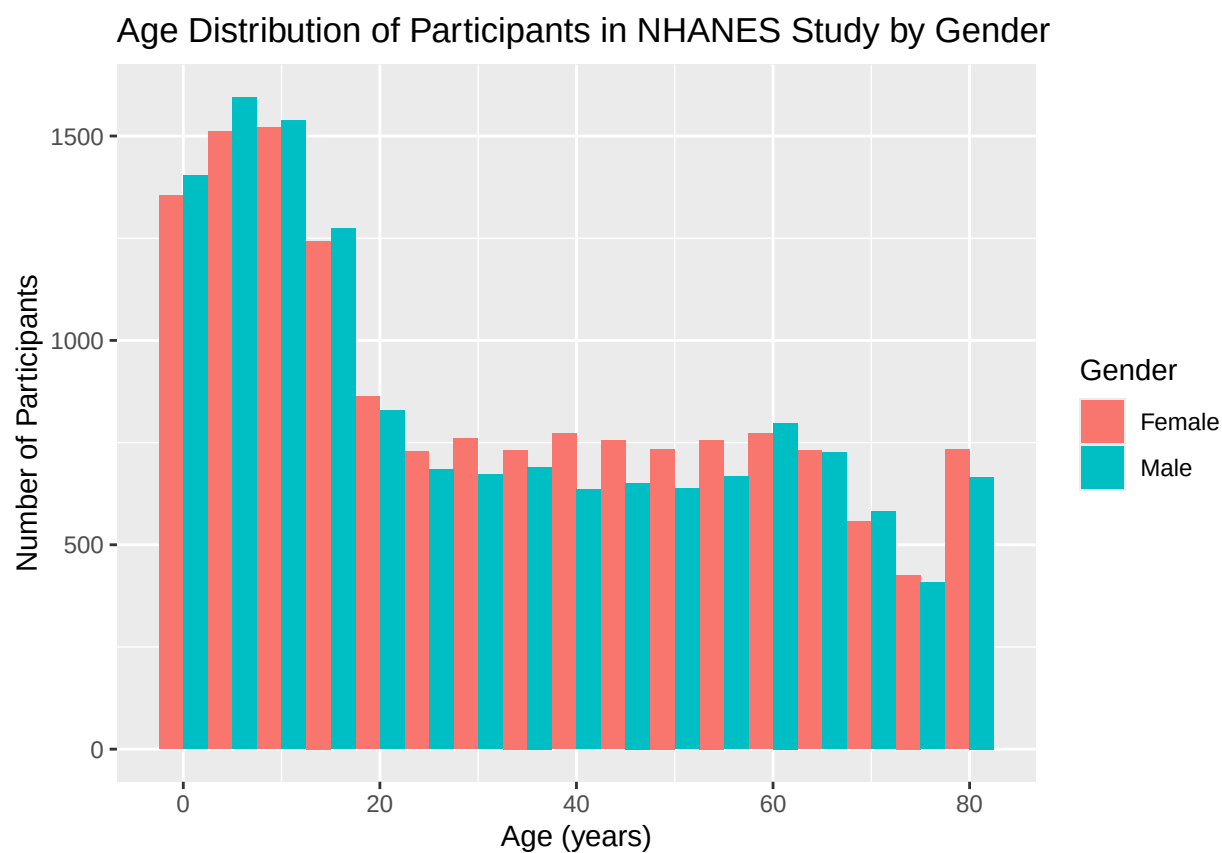
2026-09-27

## Exercise 1: Reproducing and arranging ggplot2 figures

Using the `names()` function we can determine the names of the variables in the data set. This will allow us to know what names to use for our histogram

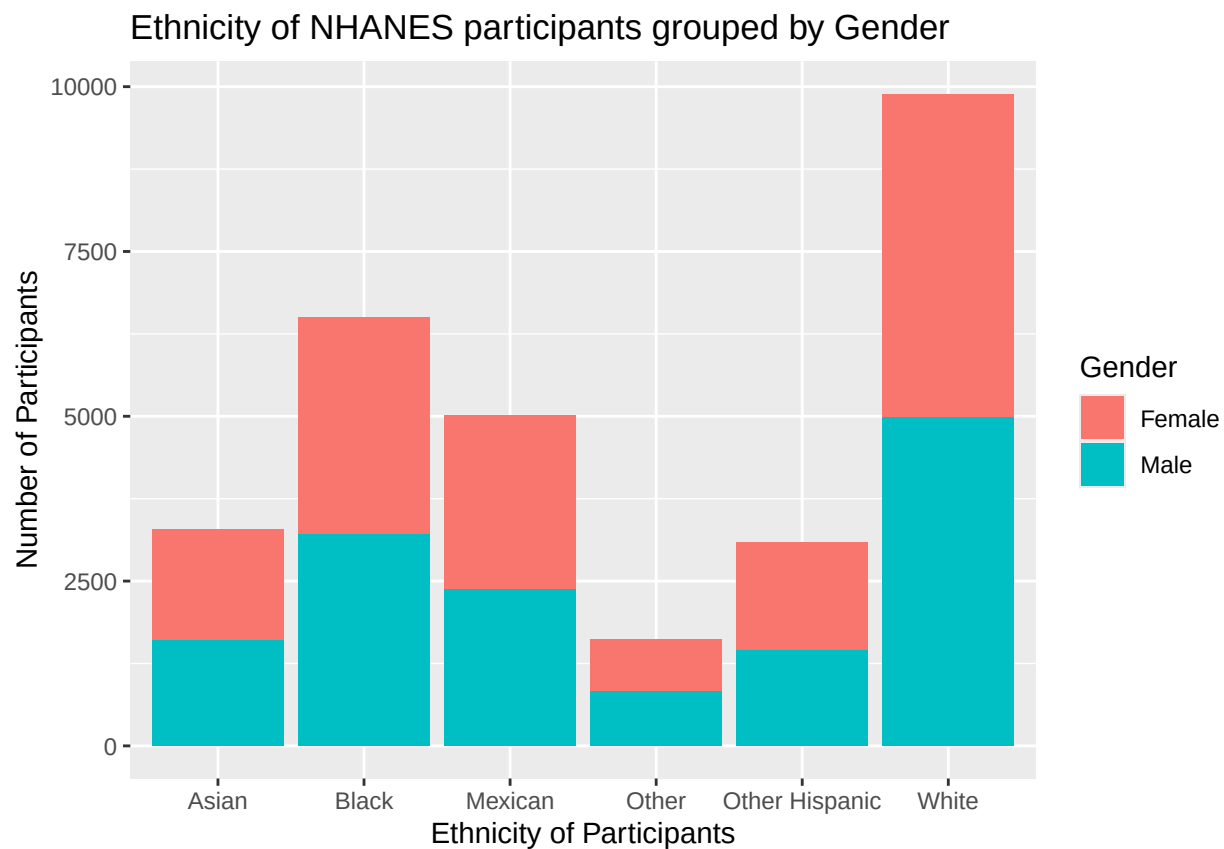
Based on the output: `RIAGENDR` has been renamed to `Gender` `RIDRETH3` has been renamed to `ethnicity_2` `RIDAGEYR` has been renamed to `age`

### Recreating the histogram using updated names



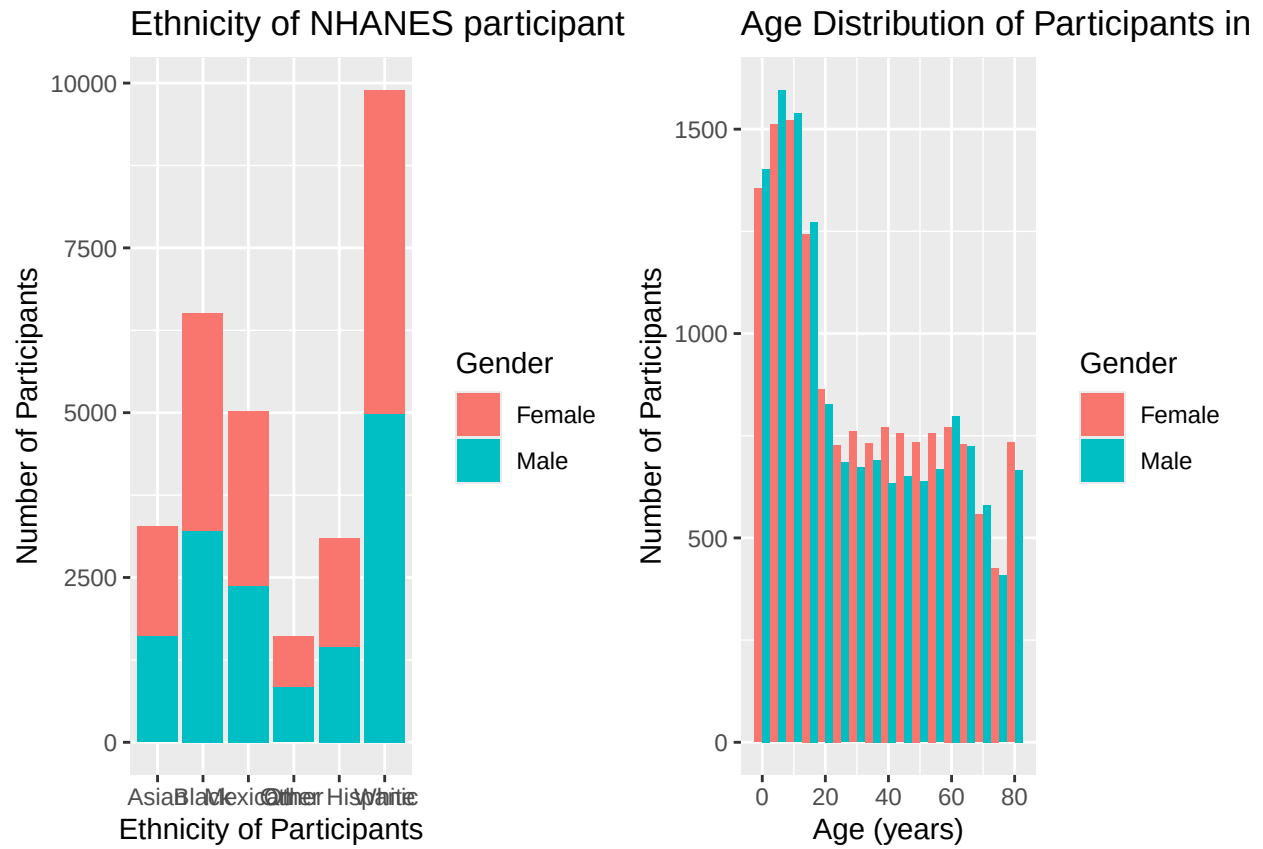
## Recreating grouped bar chart using updated names

Since the NHANES data set is not already aggregated, I used the `geom_bar()` function over the `geom_col()` function.

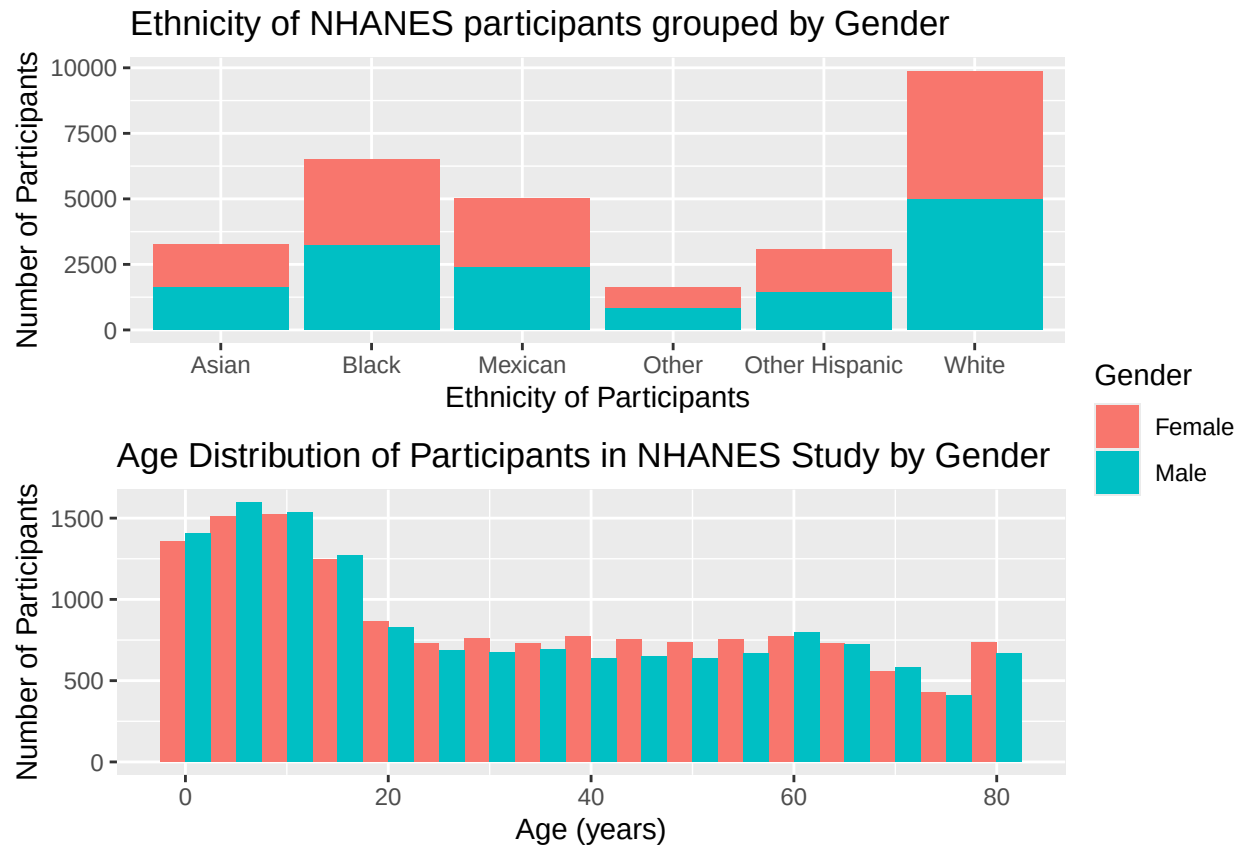


## Combining histogram and bar chart

Using the `plot_grid` function under the `cowplot` package the following figure is the output:



Using the `ggarrange` function the following figure is the output:



## Comparing the figures provided by the `ggarrange` and `plot_grid` functions:

Using the `ggarrange()` function allows you to configure and present your plots in a more organized fashion. For example, as seen in the figure created by `ggarrange()`, this function allows you to create one legend instead of having the same legend twice. Additionally, `ggarrange()` function allows you to decide placement of your figures, as seen I chose to have them vertical instead of next to each other horizontally.

## Exercise 2: Visualizing key characteristics in the NHANES data set

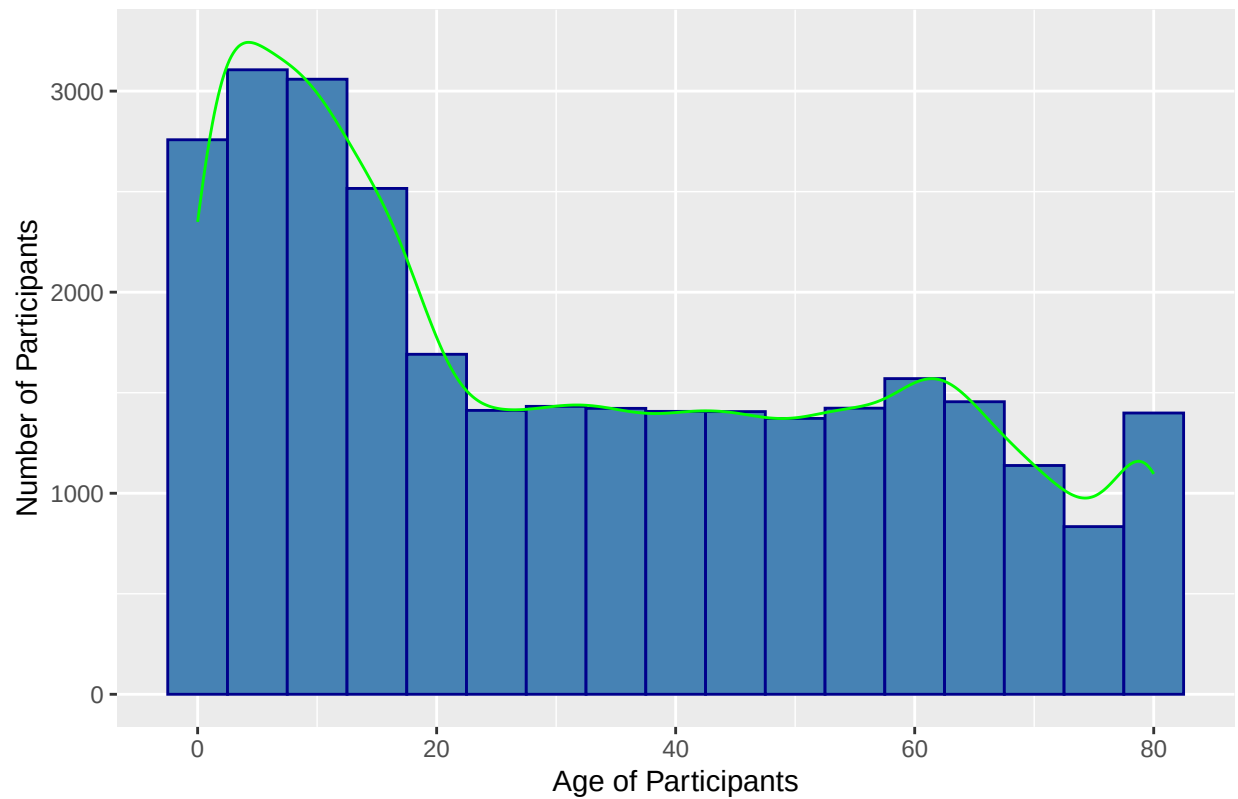
### Creating figures to represent Age

Age is represented in two ways in the NHANES cleaned data set; one as numerical and one as a categorical variable.

### Histogram and density plot to represent Age

Considering the continuous numerical variable **Age**, a histogram and density plot is the most appropriate figure. The histogram will allow you to identify which ages participated in the study more, and the density plot allows you to see if the age is skewed in a certain direction, the peak, and the overall spread of the data.

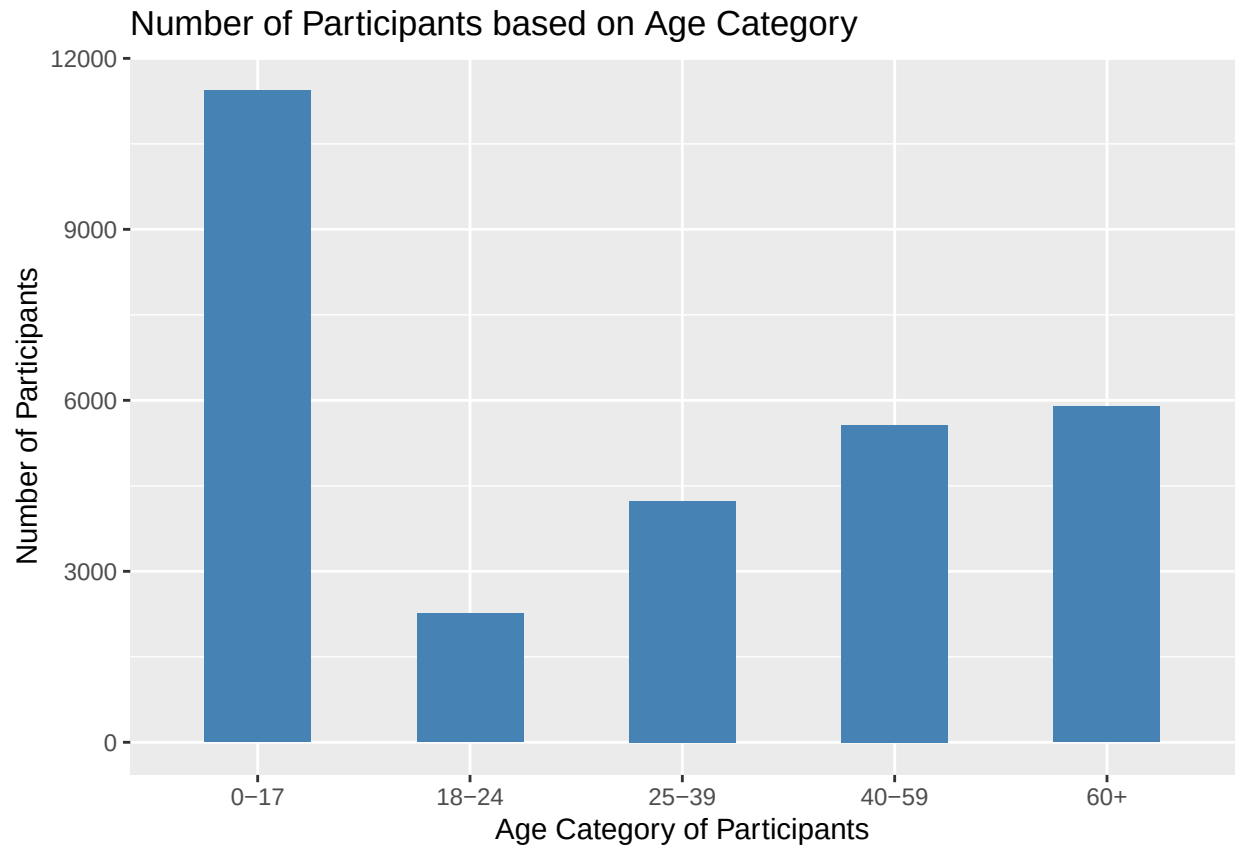
Distribution of Age of NHANES participants



To create an appropriate figure to represent the age distribution, I represent y-axis as the number of participants, I used `geom_density` to show the general skew and peak. Based on the figure produced, most participants are younger than 20.

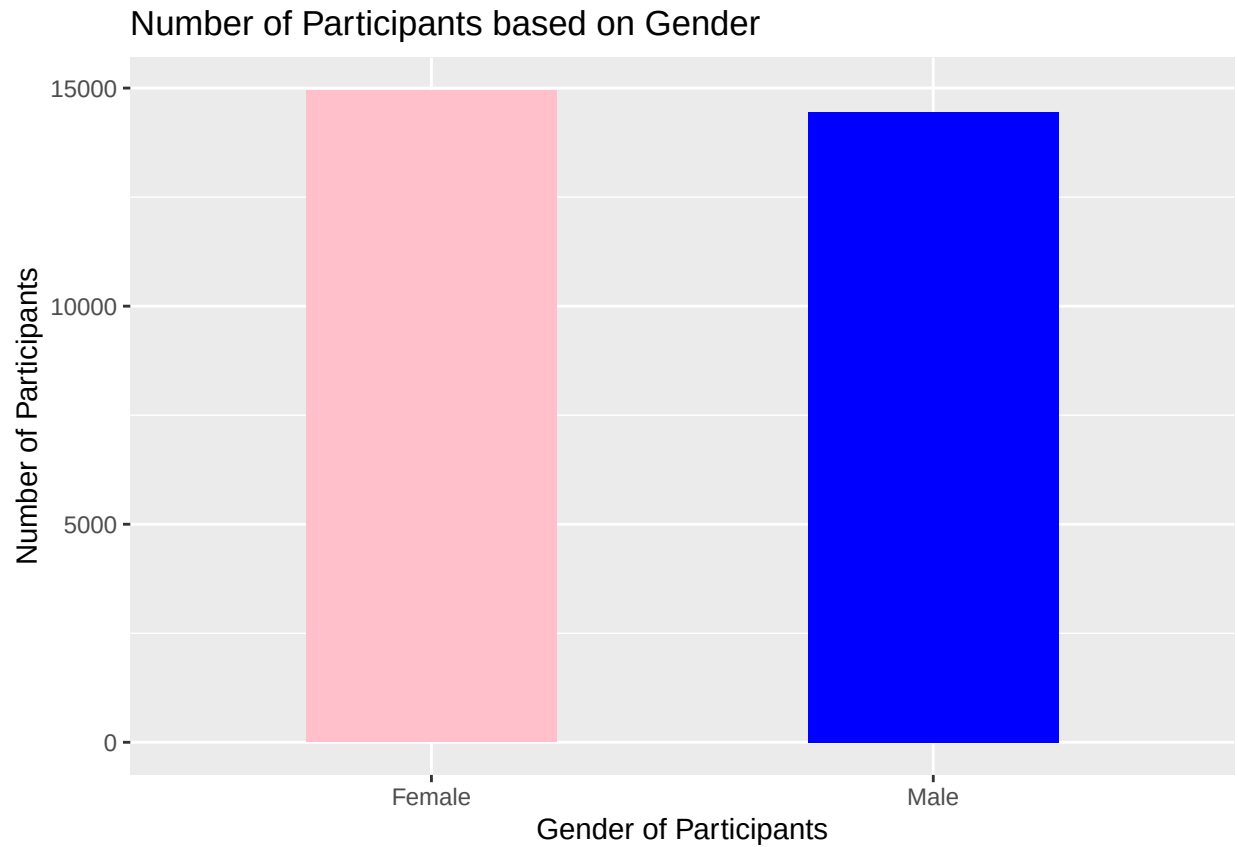
#### Bar chart to represent Age

A bar chart for the categorized age groups is most appropriate as it can allow you to compare the different age groups easier.



### Creating a bar chart to represent gender

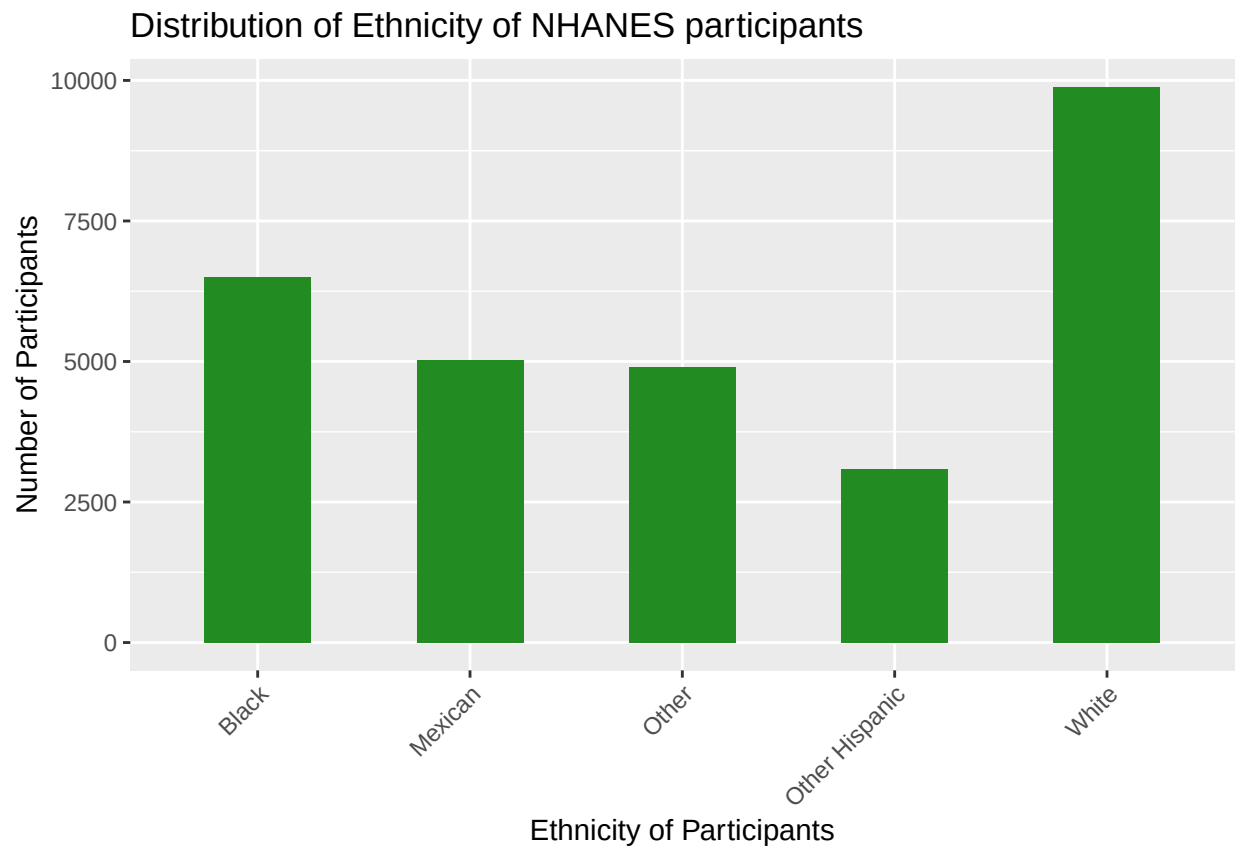
A bar chart is the most appropriate for representing the Gender variable as it a categorical variable.



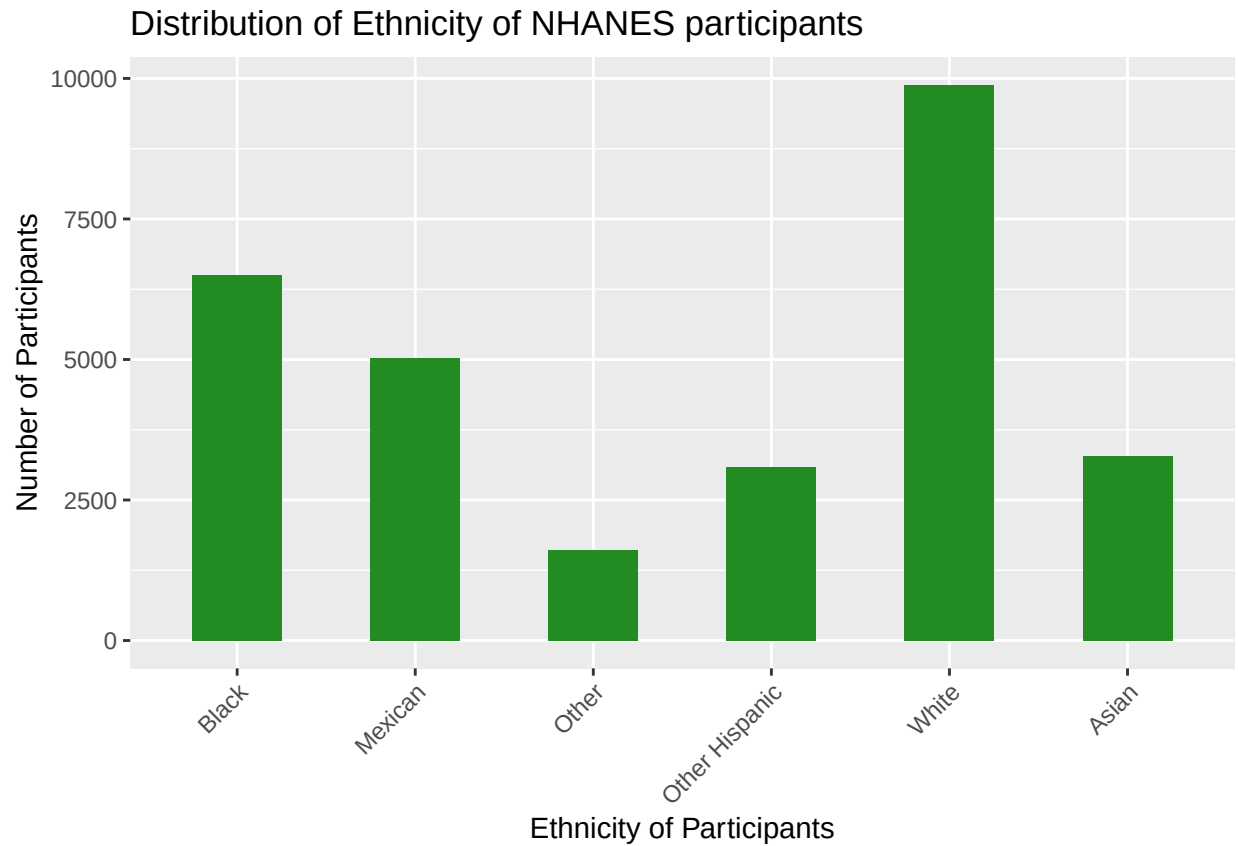
### Creating a bar chart to represent ethnicity

A bar chart is the most appropriate for representing the Ethnicity variables because it is categorical data making it easier to compare the number of participants in each distinct ethnicity.

Bar chart using Ethnicity\_1 variable



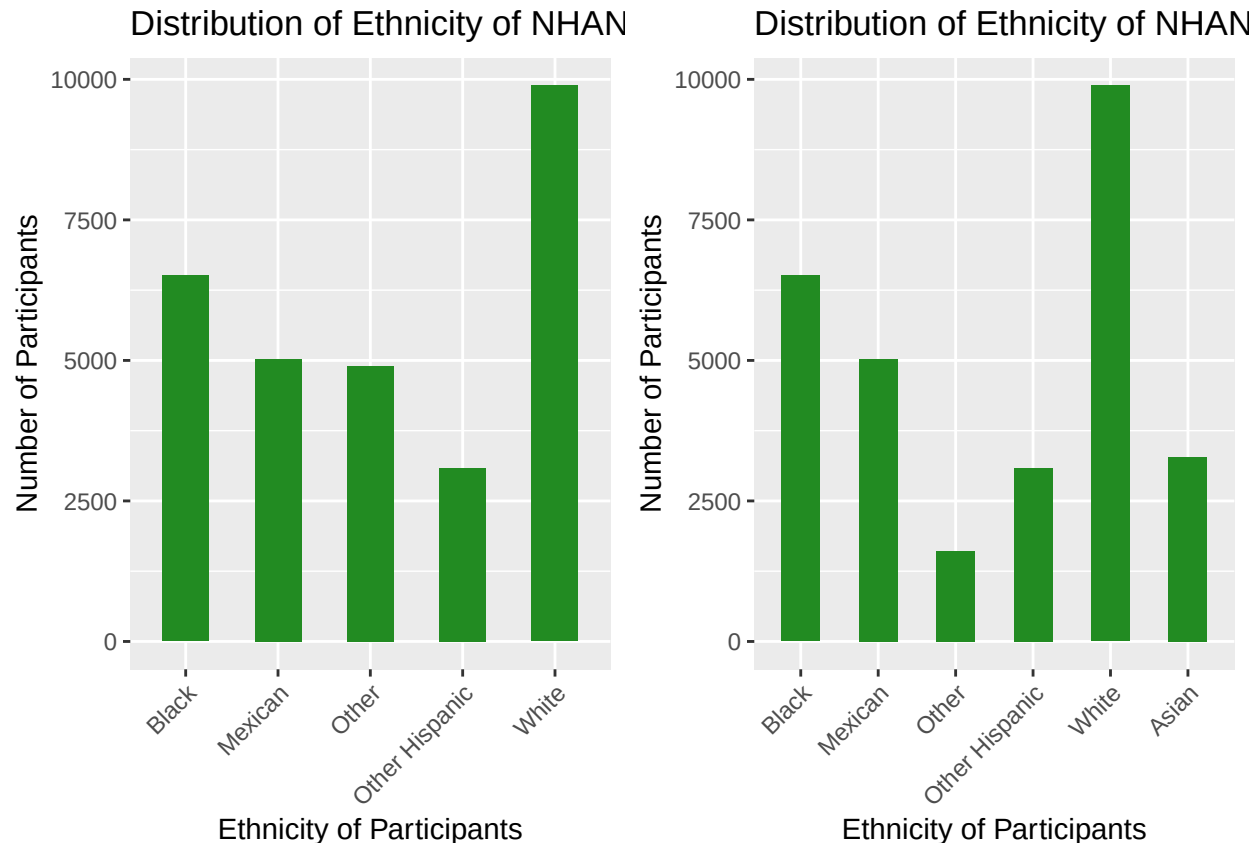
Bar chart using Ethnicity\_2 variable



Using the `levels()` function I moved the `Asian` category to the left hand side so it is easier to compare the figures.

### Comparing the two ethnicity variables

Using the `ggarrange` function allows you to compare the two charts side by side.



Based on this figure, The bar chart created using the data from `ethnicity_2` variable is more appropriate and should be chosen to keep. This is because it divides the participants into more specific categories, aka it includes `Asian` as an ethnicity instead of it all being in `Other` or `Other Hispanic`.

### Exercise 3: Improving ggplot figure

The colleagues figure is difficult to interpret as a reader who knows nothing about the study. Issues include:

1. The legend has no title explaining that the numbers represent the participant id numbers

Improvement: Using the `labeller()` function allowed me to change the title to participant id number, so it is more descriptive.

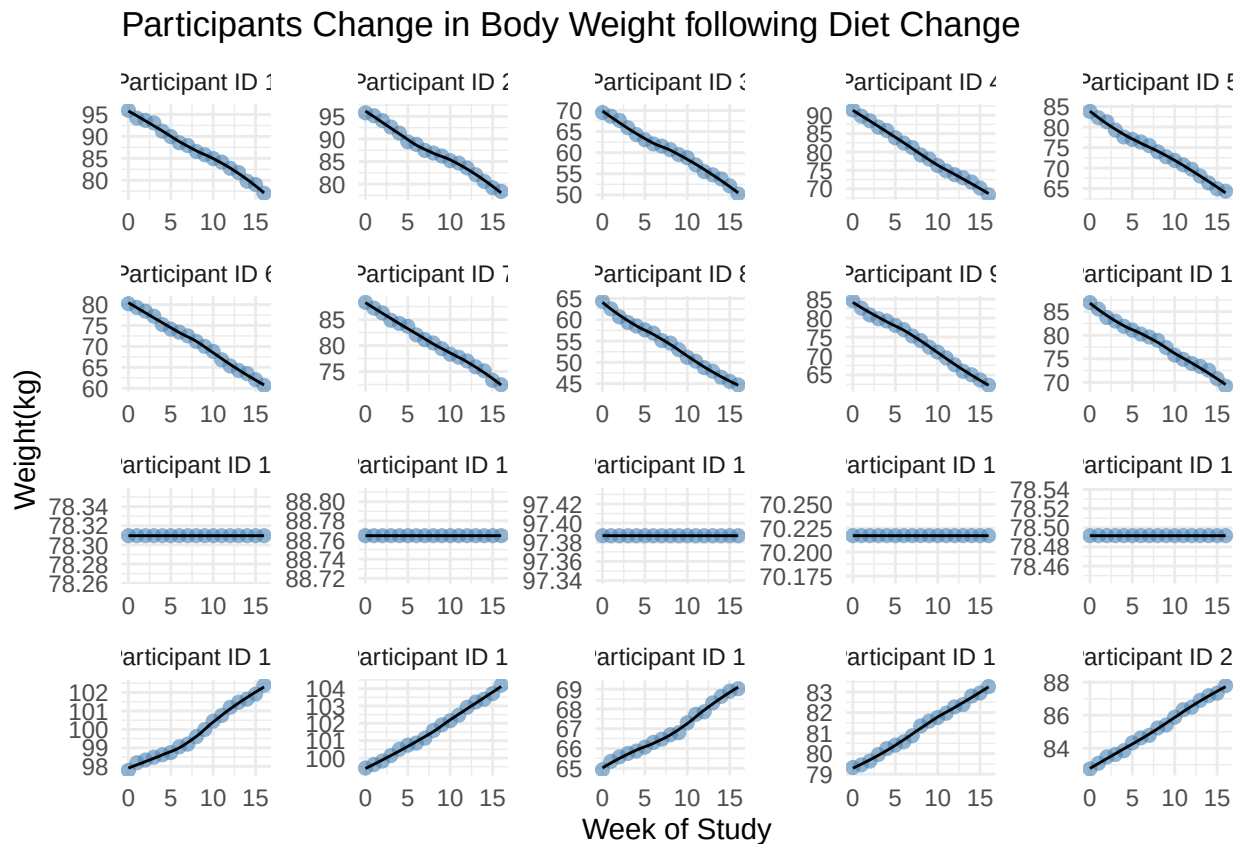
2. Y-axis title: The unit used for weight is not provided
3. X-axis title: title should be more descriptive
4. There is no title explaining the figure
5. While the progression of each participant throughout the study is shown, it is not a clear depiction of each participants trajectory.

Improvement: A) A scatter plot would be more appropriate as you can represent a participants weight each week with an individual measurement using the `geom_point()` allows you to see the individual point. The function `smooth()` also allow you to see the overall trend in the participants change in weight over the study time.

- B) Creating a figure that maps each participants change in weight throughout the study and then using the `facet_wrap()` function would be more appropriate.

Using the `names()` and `str()` function helps determine the names of the variables and the variables provided.

Using faceting allows you to see each participants weight trajectory throughout the study. This makes it easier to compare the different participants based on the change in their weight over the study. I used the `facet_wrap()` function as I am grouping it based on the single variable `participant`



## Exercise 4: Exploring relationships through visualization

To calculate the average of systolic blood pressure for each participant, the function `rowMeans()` was used. Specifically, it was used to be able to calculate the mean of each participant and their average systolic reading.

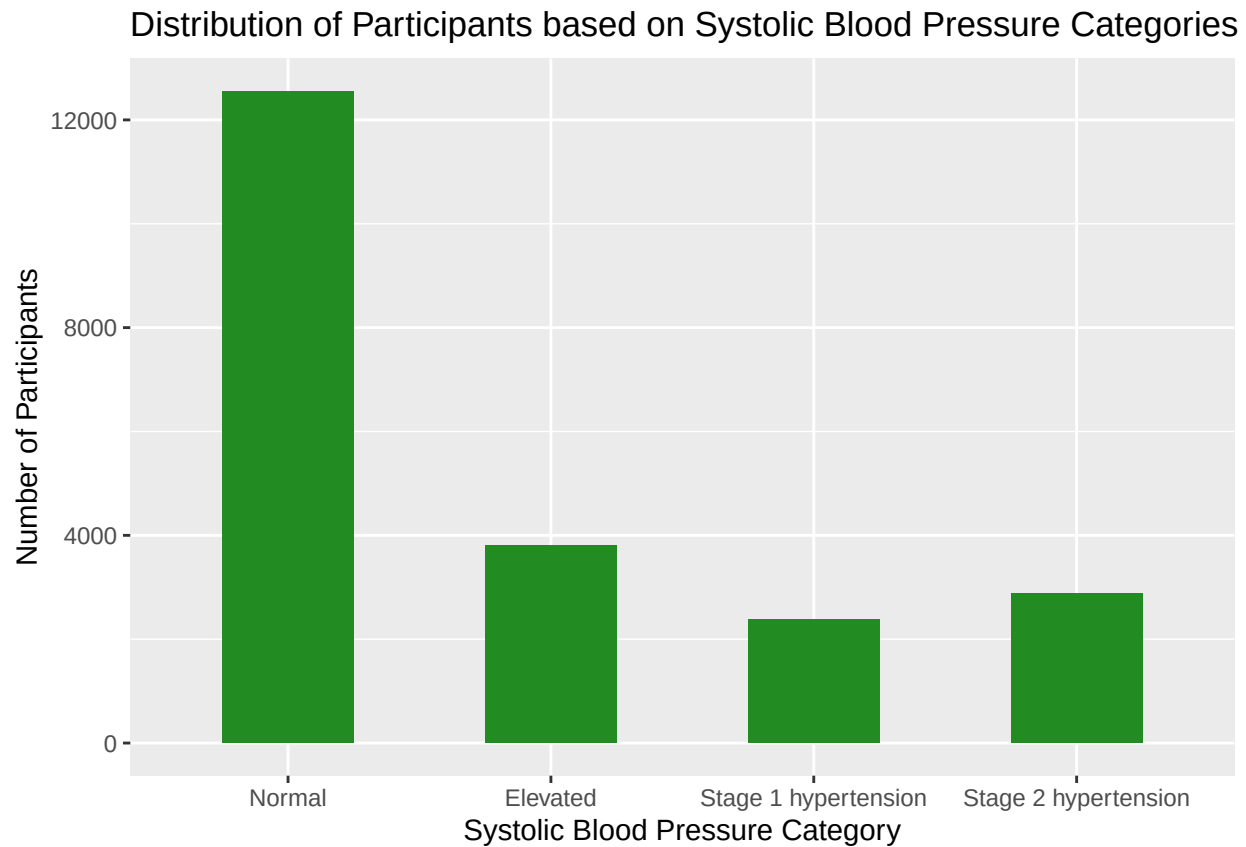
To then delete participants who had missing values for all systolic readings the `complete.cases()` function. This allows for a better picture of the data set.

After filtering the data set for missing values, to be able to visualize the distribution of participants based on their average systolic readings, categories were created using the `cut()` function. Categories were divided as “Normal” - 0-120 “Elevated” - 120-130 “Stage 1 hypertension” - 130-140 “Stage 2 hypertension” - 140-250

To actually visualize the new data final data set, figures were created to understand the distribution of participants based on their age, gender, their systolic blood pressure category.

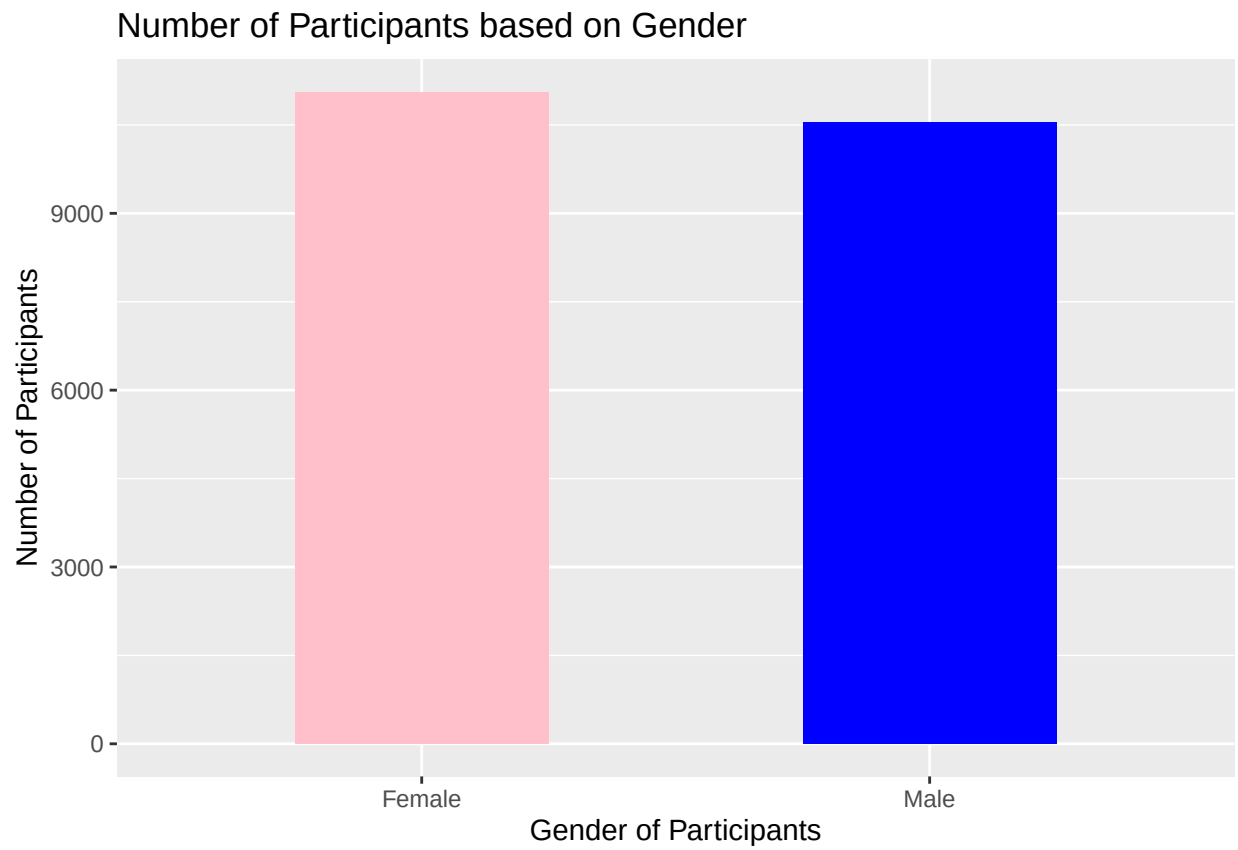
## Representing Final Systolic Blood Pressure Data

Using the `geom_bar()` I used a bar chart to represent the final systolic blood pressure categories of participants.



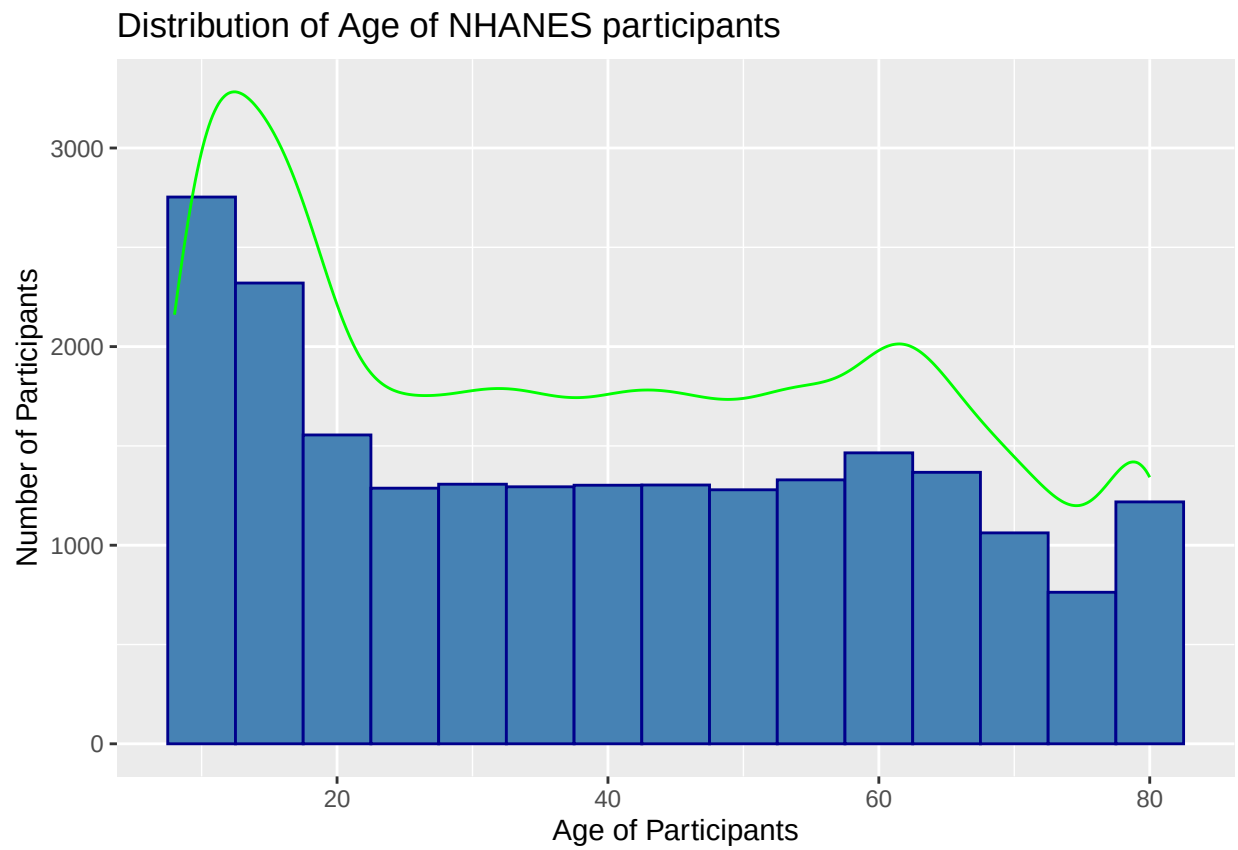
Based on this graph the majority of participants have normal systolic blood pressure. It is interesting to note, more people have stage 2 hypertension than stage 1.

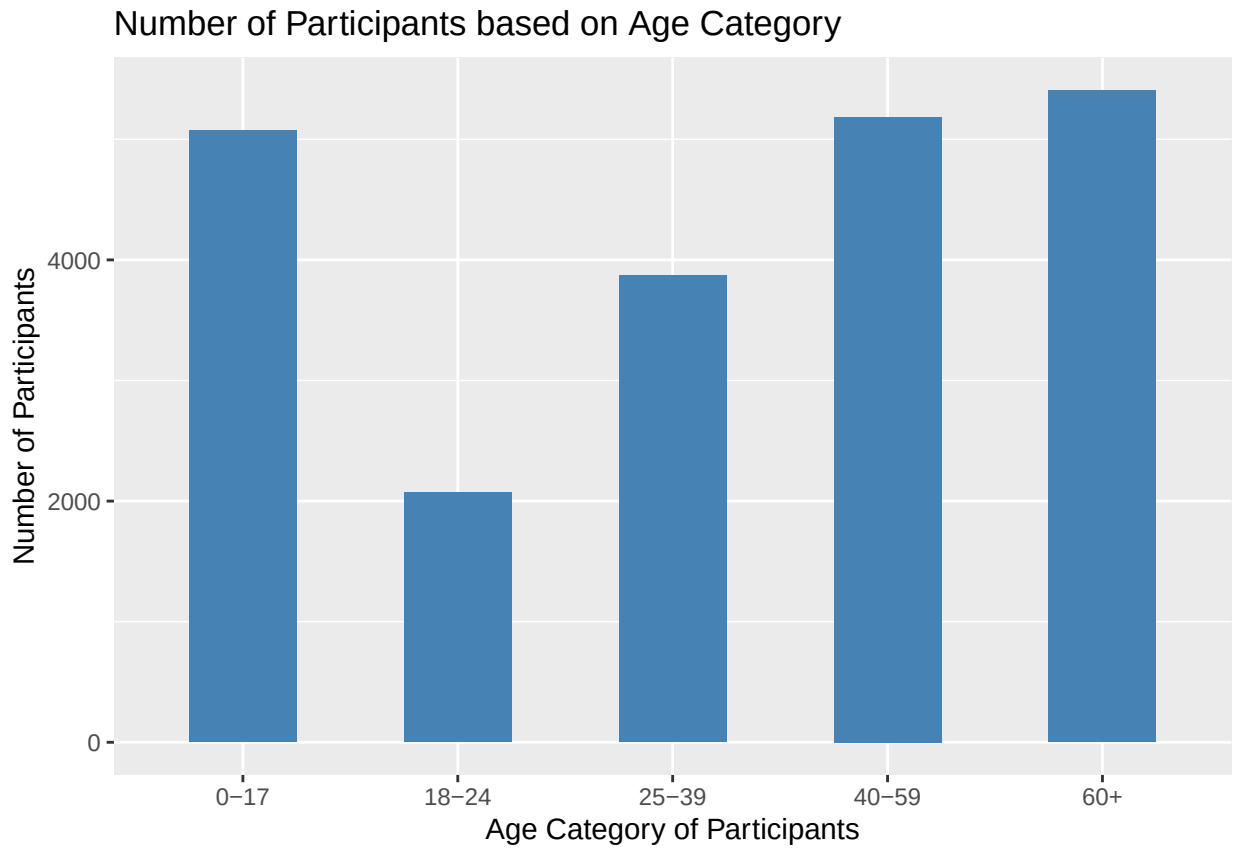
## Representing Gender Distribution in Final data set



Based on this distribution, there are marginally more females than males that participated in this study.

## Representing Final Distrubution of Age of Participants





Step 5

## Difference in Systolic Blood Pressure Categories based on Age and Gender

